Perfect difference sets: i.o. lower bounds, exact small-N optima, and an extension obstruction

Notes on Erdős Problem #1194

May 11, 2026

Abstract

A set $A \subset \mathbb{N}$ is a perfect difference set (PDS) if every $n \geq 1$ admits a unique representation $n = a_n - b_n$ with $a_n, b_n \in A$. Erdős asked how fast a_n/n must increase [1]. We give three contributions. First, we record full proofs of the infinitely-often lower bounds $a_n \gg n \log n$ (classical), $a_n \gg n^{2-o(1)}$ (GPT-5.4 Pro / Bloom, April 2026), and the logarithmic refinement $a_n \gg n^2/f(n)$ for any nondecreasing slowly varying f with $\sum 1/(nf(n)) < \infty$. Second, we compute exactly the value $M^*(N) := \min\{\max(A) : A \text{ Sidon}, \ [1, N] \subset A - A\}$ for $N \leq 45$ via branch-and-bound and CP-SAT, observing $M^*(N) \approx 2N$. Third, we exhibit an empirical *extension obstruction*: the optimum $A^*_{N_0}$ for the finite problem is a poor starting seed for an infinite PDS, with forced extensions growing like N^{4+}, worse than Lev's greedy. We formulate a corresponding "for all large n" conjecture and discuss why the partition-identity argument is sharp at the $n^{2-o(1)}$ ceiling. Code and data accompanying this note live in `erdos_1194/`.

Contents

1 Setup and notation

A set $A \subset \mathbb{N}$ is a *perfect difference set* (PDS) if every $n \geq 1$ admits a *unique* representation

$$n = a_n - b_n, \qquad a_n, b_n \in A, \quad a_n > b_n.$$

The notation a_n is fixed: it is the larger element of the unique representing pair for n. It is *not* the n-th element of A. To avoid confusion we write the increasing enumeration of A as

$$A = \{x_1 < x_2 < x_3 < \cdots\}.$$

A PDS is in particular a Sidon set: all pairwise differences are distinct. We write $A_x := A \cap [1, x]$, $k(x) := |A_x|$, and $N(x) := \#\{n \geq 1 : a_n \leq x\}$. The running maximum is $M(N) := \max_{n \leq N} a_n = \max(\bigcup_{n \leq N}\{a_n, b_n\})$ when A is built greedily from $n = 1$ outward.

2 The PDS counting identity

Lemma 1 (PDS counting)**.** *For every $x \geq 1$, $N(x) = \binom{k(x)}{2}$.*

Proof. Each $n \leq N(x)$ has $a_n \leq x$, hence both $a_n, b_n \in A_x$. The map $n \mapsto \{a_n, b_n\}$ is injective into unordered pairs of A_x, so $N(x) \leq \binom{k(x)}{2}$. Conversely, every unordered pair $\{a, b\} \subset A_x$, $a > b$, gives a difference $a - b \in [1, x-1]$ which by uniqueness is the representation of some n with $a_n = a$, $b_n = b$, $a_n \leq x$. Hence $\binom{k(x)}{2} \leq N(x)$. $\square$

This identity will be used repeatedly. As a corollary, $\binom{k(x)}{2} \leq x$, recovering the standard Sidon density bound $k(x) \leq (1 + o(1))\sqrt{2x}$ as an inequality only — but PDS gives an *equality* $N(x) = \binom{k(x)}{2}$, not merely an inequality.

3 Infinitely-often lower bounds

3.1 The classical $n \log n$ bound

Theorem 1 (Erdős; classical)**.** *For any infinite PDS A, $a_n \gg n \log n$ for infinitely many n.*

Proof. By Erdős's i.o. Sidon density bound (see e.g. Halberstam–Roth [2], Ch. II), there exist $x_j \to \infty$ with $k(x_j) \leq C\sqrt{x_j / \log x_j}$. By Lemma 1, $N(x_j) = \binom{k(x_j)}{2} \leq C^2 x_j / (2 \log x_j)$. Setting $n_j := N(x_j) + 1$, the element $a_{n_j} > x_j$ and $n_j \leq C^2 x_j / (2 \log x_j) + 1$, so

$$a_{n_j} > x_j \geq \frac{2 \log x_j}{C^2}\,(n_j - 1) \geq \frac{2(1 + o(1))}{C^2}\, n_j \log n_j. \qquad \square$$

3.2 The partition identity and $n^{2-o(1)}$

The next theorem improves Theorem 1 to almost-n^2 via a completely different argument due to GPT-5.4 Pro and condensed by T. Bloom in the comment thread of erdosproblems.com/1194 (post 5745, April 23 2026). The argument does not use Sidon density at all; it uses the following partition identity together with a gap lower bound.

Lemma 2 (PDS partition)**.** *For each $k \geq 2$ let $D_k := \{x_k - x_i : 1 \leq i < k\}$. Then $\{D_k\}_{k\geq 2}$ is a partition of $\mathbb{Z}_{>0}$. Consequently, for every $X \geq 1$,*

$$X = \sum_{k\geq 2} |D_k \cap [1, X]|.$$

Proof. Every $n \geq 1$ has a unique representation $n = a_n - b_n$. Pick k with $a_n = x_k$ and i with $b_n = x_i$; then $n = x_k - x_i \in D_k$. Uniqueness of the representation gives uniqueness of k. □

Lemma 3 (Gap bound from $a_n \ll n^c$)**.** *Suppose $a_n \leq C\, n^c$ for all $n \geq 1$ and some $c > 1$, $C > 0$. Then $x_k - x_{k-1} \geq (x_k/C)^{1/c}$ for every $k \geq 2$, and consequently $x_k \gg k^{c/(c-1)}$.*

Proof. Let $d_k := x_k - x_{k-1}$. The pair (x_k, x_{k-1}) realises difference d_k, so by PDS uniqueness $a_{d_k} = x_k$. Hence $x_k \leq C d_k^c$ and $d_k \geq (x_k/C)^{1/c}$. Telescoping, $x_k \geq \sum_{i=2}^{k} d_i \geq C^{-1/c} \sum_i x_i^{1/c}$, which is dominated by $x_k \gg k^{c/(c-1)}$ by an elementary comparison with the ODE $dy/dk = c_0 y^{1/c}$. □

Theorem 2 (GPT-5.4 Pro, after Bloom)**.** *For any infinite PDS A and any $c \in (1, 2)$, the inequality $a_n \leq C\, n^c$ for all n is impossible (for any C). Equivalently,* $\limsup_{n\to\infty} a_n/n^c = \infty$ *for every $c < 2$, i.e. $a_n \gg n^{2-o(1)}$ infinitely often.*

Proof. Suppose, for contradiction, that $a_n \leq C\, n^c$ for all n and some $c \in (1, 2)$. By Lemma 3, $x_k \gg k^{c/(c-1)}$.

Fix $X \geq 1$ and let $K = \lfloor X^{(c-1)/c} \rfloor$. By Lemma 2,

$$X = \underbrace{\sum_{2\leq k\leq K} |D_k \cap [1, X]|}_{\text{(I)}} + \underbrace{\sum_{k>K} |D_k \cap [1, X]|}_{\text{(II)}}.$$

For (I), trivially $|D_k| \leq k - 1$, so (I) $\leq K^2/2 \ll X^{2-2/c}$.

For (II), $k > K$ gives $x_k > 2X$ for X large (absorb a constant into K). Any consecutive $x_{j-1} < x_j$ in $A \cap [x_k - X, x_k)$ satisfies $x_j - x_{j-1} \geq (x_j/C)^{1/c} \gg x_k^{1/c}$ by Lemma 3 and $x_j \geq x_k/2$. Hence $|D_k \cap [1, X]| \leq X/x_k^{1/c} \ll X \cdot k^{-1/(c-1)}$. Since $1/(c-1) > 1$, the sum $\sum_{k>K} k^{-1/(c-1)}$ converges with tail $\ll K^{(c-2)/(c-1)}$, giving (II) $\ll X \cdot K^{(c-2)/(c-1)} \ll X^{2-2/c}$.

Combining, $X \ll X^{2-2/c}$, i.e. $X^{2/c-1} \ll 1$ uniformly. Since $c < 2$, $2/c - 1 > 0$ and the left side $\to \infty$. Contradiction. □

3.3 Logarithmic refinement

The boundary $c = 2$ in Theorem 2 is sharp; the same scheme under the slightly weaker hypothesis $a_n \leq Cn^2/f(n)$ pushes the bound to the convergence boundary of $\sum 1/(nf(n))$.

Theorem 3 (Logarithmic refinement)**.** *Let $f\colon \mathbb{N} \to \mathbb{R}_{>0}$ be nondecreasing, slowly varying (i.e. $f(\lambda n)/f(n) \to 1$ for every fixed $\lambda > 0$), unbounded, with $\sum_{n\geq 2} \frac{1}{nf(n)} < \infty$. Then for any infinite PDS A,*

$$a_n \gg \frac{n^2}{f(n)} \qquad \text{for infinitely many } n.$$

In particular $a_n \gg n^2/(\log n)^{1+\varepsilon}$ i.o. for every $\varepsilon > 0$, and $a_n \gg (n/\log n)^2$ i.o.

Proof sketch. Replace $a_n \leq Cn^c$ by $a_n \leq Cn^2/f(n)$. The gap bound becomes $d_k \geq \sqrt{x_k f(d_k)/C}$, and for slowly varying f this iterates to $x_k \gg k^2 f(k)$. The anchor window bound in Step (II) becomes $|D_k \cap [1, X]| \leq X/(kf(k))$, and the partition sum split at $K = \Theta(\sqrt{X/f(\sqrt{X})})$ gives $X \leq K^2/2 + X\varepsilon_K + o(X)$ where $\varepsilon_K = \sum_{k>K} 1/(kf(k)) \to 0$ by the convergence hypothesis. Since $K^2 = o(X)$ (from $f(\sqrt{X}) \to \infty$) and $\varepsilon_K \to 0$, we obtain $X \leq 3X/4$ for X large — a contradiction. (Full proof in `lower_bound_io.tex` of the companion repository.) □

Remark. *The partition-identity argument is sharp at the convergence boundary $\sum 1/(nf(n)) < \infty$: at $f(n) = \log n$ (where the sum diverges), the tail ε_K no longer goes to zero with K and the contradiction collapses. Pushing past requires either a sharper density estimate in the anchor window or a non-partition counting identity. A computational probe (`fourier_probes.py`) on greedy data verifies the Fejér PDS identity $\int_0^1 K_T(\xi)|\widehat{1_A}(\xi)|^2 d\xi = |A| + T - 1$ exactly, and shows the anchor window count is uniformly $\approx 0.42 \times |A|/\sqrt{\max A}$ — a constant multiple of greedy's overall Sidon density, with no exploitable PDS-specific slack.*

Remark (*Typographical note*)**.** *At the time of writing, `erdosproblems.com/1194` states the conclusion of Theorem 3 under the condition $\sum 1/(nf(n))$* diverges. *This appears to be a copy-edit error: the proof requires convergence, and so does Bloom's example $f(n) = (\log n)^2$ in the comment thread (post 5784).*

4 Exact small-N optima

4.1 The finite problem

For each $N \geq 1$, let

$$M^*(N) \;:=\; \min\bigl\{\max(A) : A \subset \mathbb{N},\ A \text{ Sidon},\ \{1, M\} \subset A,\ [1, N] \subset A - A\bigr\}.$$

This is the minimum value of $\max(A)$ over all finite Sidon sets that realise every difference in $[1, N]$ at least once. Trivially $M^*(N) \geq N + 1$ and $M^*(N) \geq 1 + \binom{K}{2}$ where $K = \lceil (1 + \sqrt{1 + 8N})/2 \rceil$ is the smallest K with $\binom{K}{2} \geq N$.

4.2 Exact values

We computed $M^*(N)$ via two engines: a Python bitmask depth-first search (`exact_search.py`) and a Google ortools CP-SAT model (`exact_search_cpsat.py`). For each N, both the feasibility at $M^*(N)$ and the infeasibility at $M^*(N)-1$ are verified exhaustively. Lev's greedy PDS (`explore.py`) gives an upper reference.

Proposition 1 (Empirical scaling)**.** *Across the data of Table 1, $M^*(N)/N \in [1.60, 2.15]$, hovering close to the trivial floor $N + 1$.*

N	M^* exact	greedy max A	gd / M^*	$\lvert A^*\rvert$	$\lvert A_{\text{gr}}\rvert$
20	36	97	2.7×	8	11
25	46	225	4.9×	9	15
30	56	444	7.9×	10	19
35	56	625	11.2×	10	21
40	86	1175	13.7×	12	27
45	86	1492	17.4×	12	29
50	≥ 92	2518	≥ 27.4×	—	35

Table 1: Finite PDS optima vs greedy. M^* verified to optimality for $N \leq 45$; $N = 50$ proves $M^* > 91$ (CP-SAT exhausts at $M = 91$ infeasible, $M = 92$ times out at 35 s/M).

The empirical conclusion is $M^*(N) \approx 2N$. By Sidon density, $|A^*| \approx \sqrt{2N}$, and indeed Table 1 shows $|A^*|$ within 1 of $\lceil (1 + \sqrt{1 + 8N})/2 \rceil$.

Witness sets (sorted) for the first few N:

- $N = 20$: $\{1, 5, 6, 18, 20, 26, 29, 36\}$.
- $N = 30$: $\{1, 2, 7, 11, 24, 27, 35, 42, 54, 56\}$.
- $N = 40$: $\{1, 10, 11, 18, 31, 43, 46, 57, 62, 80, 84, 86\}$.

5 The extension obstruction

The finite optima of Section 4 are extremal: they pack $\binom{|A^*|}{2}$ distinct differences into roughly $[1, 2N]$, saturating Sidon density. We now show empirically that these extremal seeds are *terrible starting points* for extending to a larger PDS.

5.1 Extension cost

For a seed A_{seed} (a Sidon subset of $\mathbb{N}$), define

$$M_{\text{ext}}(A_{\text{seed}}, N) \;:=\; \min\bigl\{\max(A) : A \supseteq A_{\text{seed}},\ A \text{ Sidon},\ [1, N] \subset A - A\bigr\}.$$

Each step is again solved to optimality via CP-SAT (`extension_cost.py`). Two trajectories were computed:

N	M_{ext} from A^*_{20}	M_{ext} from A^*_{30}	greedy	unconstrained M^*
20	36	—	97	36
30	144	56	444	56
40	344	168	1175	86
45	1054	254	1492	86
56	—	516	3358	—
60	—	1019	—	—
63	—	1590	—	—

Table 2: Extension cost from finite optima vs scratch greedy vs unconstrained M^*.

Log-log fits on the "hard" steps (those requiring new elements) give:

- Seed A^*_{20}: $M_{\text{ext}} \sim N^{3.80}$ overall, $N^{6.48}$ in the tail.
- Seed A^*_{30}: $M_{\text{ext}} \sim N^{4.27}$ overall, $N^{4.90}$ in the tail.

5.2 Interpretation

We summarise the chain:

$$M^*_{\text{unconstrained}}(N) \sim N \quad \ll \quad M_{\text{greedy}}(N) \sim N^3 \quad \ll \quad M_{\text{ext, seeded}}(N) \gtrsim N^4. \tag{1}$$

A finite extremal optimum $A^*_{N_0}$ packs $\binom{|A|}{2}$ distinct differences into a window of width $\sim 2N_0$, leaving very few free difference slots. Every subsequent element added must avoid a densely packed set of forbidden differences. Greedy avoids this trap by being deliberately conservative (every Strategy-2 element lives at $\geq \max(A) + n + 1$), keeping the difference budget loose at the cost of a suboptimal finite $\max(A)$.

5.3 A conjecture for the infinite problem

The chain (1) is consistent with the i.o. lower bound $a_n \gg n^{2-o(1)}$ of Theorem 2, and in fact strengthens it empirically: a *naive* gluing of finite optima produces a_n growth strictly worse than N^3. Beating Lev's greedy requires a construction that balances tightness at current N against future extensibility — a non-local property.

This motivates the following "for all n" analog:

Conjecture 1 (Extension lower bound)**.** *There exists $c \in (1, 3]$ such that for every infinite PDS A, $a_n \gg n^c$ for all sufficiently large n. The value of c is governed by the asymptotic exponent of $M_{ext}(A_0, N)$ as the seed A_0 ranges over Sidon sets with* $\max(A_0) \to \infty$.

Conjecture 1 is strictly stronger than the i.o. form of Theorems 2 and 3. Empirically the extension-cost trajectories suggest c is a moderate constant (somewhere between 2 and 3), but the data does not distinguish a specific value.

6 Constructive upper bound and the cubic ceiling

6.1 Lev's greedy and its constant

Lev [3] proves that the greedy construction (always choose the smallest valid element to cover the next missing difference) produces an infinite PDS with $a_n \leq Cn^3$ for some $C > 0$. Empirically [5] we observe

$$\frac{\max_{n \leq N} a_n}{N^3} \;\to\; 0.0293 \pm 0.0006 \qquad (N \in [1500, 1900]).$$

6.2 Dense-Sidon-seeded greedy

Replacing the empty initial A with a known dense Sidon set (Mian–Chowla, Erdős–Turán, or a Phase 4 optimum) and running the same Strategy 1 / Strategy 2 greedy as Lev's gives a strictly better constant:

Three observations from Table 3:

1. Mian–Chowla seeding reduces the empirical constant from $\approx 0.025\, N^3$ to $\approx 0.0023\, N^3$.

seed	$\lvert A_{\text{seed}} \rvert$	N	seeded max A	ratio $/N^3$
scratch greedy	0	500	3,080,237	0.02464
Erdős–Turán $p = 23$	23	500	544,923	0.00436
Mian–Chowla $n = 30$	30	500	368,611	0.00295
Mian–Chowla $n = 60$	60	500	286,281	0.00229
Mian–Chowla $n = 100$	100	500	423,111	0.00339
Phase 4 opt A^*_{30}	10	500	881,451	0.00705

Table 3: Dense-Sidon-seeded greedy: the constant in the N^3 scaling drops by up to 10.8× when seeded with Mian–Chowla of size 60. The Phase 4 optimum is a *bad* seed.

2. The Phase 4 optimum A^*_{30}, despite being the densest possible Sidon set covering $[1, 30]$, is a *worse* seed than Mian–Chowla — giving ratio 0.00705 vs 0.00229. This is a direct empirical witness of the extension obstruction (Section 5): tight seeds are extension traps.

3. No seed produces sub-cubic asymptotic growth. The constant moves; the $\Theta(N^3)$ scaling does not. Achieving $a_n = o(n^3)$ (Cilleruelo–Nathanson Problem 1 of [4]) requires a construction that is fundamentally non-greedy.

7 Open problems and the $c = 2$ barrier

7.1 What is sharp

Two boundaries appear sharp in our analysis:

- The partition argument is sharp at the convergence boundary $\sum 1/(nf(n)) < \infty$ (Remark 3.3). The computational probes find no slack in Step (II) beyond the Sidon-density factor.
- Greedy-style construction is sharp at $\Theta(N^3)$ across all seeds tested (Table 3). Seeding gives constant-factor improvements; the exponent is robust.

7.2 The three open questions

1. **Better i.o. bound.** Can $a_n \gg n^c$ i.o. be shown for some explicit $c > 2$? Phase 2B suggests this requires restriction / decoupling estimates or a non-partition counting identity.

2. **Cilleruelo–Nathanson Problem 1.** Does there exist a PDS with $a_n = o(n^3)$? Our Phase 3' experiments confirm the ($\Theta(N^3)$)-ceiling under any greedy-style continuation; non-greedy constructions (block-structured, probabilistic, Cilleruelo–Nathanson-style) remain to be tested.

3. **For-all-n lower bound.** Is Conjecture 1 true? Even an explicit $c > 1$ would be the first "for all large n" improvement on the trivial $a_n \geq n + 1$.

7.3 Beyond partition: candidate routes

Honest assessment of where to look:

- For (1), the Fourier-side Plancherel statement $\int |\widehat{1_A}|^4 = 2|A|^2 - |A|$ holds for any Sidon set and doesn't distinguish PDS. Higher moments (L^6, L^8) might. Bourgain–Demeter decoupling for the parabola has been used to bound Sidon sets; an analog for PDS would be new.

- For (2), the natural attack is block-structured constructions: concatenate Mian–Chowla, Singer, or Erdős–Turán blocks with carefully chosen offsets. Cross-block differences could be arranged to cover the medium-range n, leaving small-range and large-range n to within-block differences. We have not yet tested this; it is the natural Phase 6.
- For (3), one would convert the empirical Conjecture 1 into a theorem by showing: any infinite PDS truncated to $[1, M]$ produces a Sidon set A_M whose "extension obstruction" is at least polynomial. The Phase 4' data is suggestive but we do not see a path to proof.

Acknowledgements and code

This note records work done in the directory `erdos_1194/` of the accompanying repository. Specifically:

`explore.py`, `build_checkpoint.py` Lev's greedy.

`exact_search.py`, `exact_search_cpsat.py` Phase 4 finite-optimum search.

`extension_cost.py` Phase 4' extension cost.

`seeded_greedy.py` Phase 3' dense-Sidon-seeded greedy.

`fourier_probes.py` Phase 2B Fourier observables.

`verify_and_analyze.py` Phase 0 / 1 verification + stratification.

The arguments of Theorems 2 and 3 are due to GPT-5.4 Pro and T. F. Bloom (April 2026 comment thread on erdosproblems.com/1194). The remainder is original to this work.

References

[1] P. Erdős, *Some applications of Ramsey's theorem to additive number theory*, European J. Combin. 1 (1980) no. 1, 43–52.

[2] H. Halberstam and K. F. Roth, *Sequences*, Oxford University Press, 1966.

[3] V. F. Lev, *Reconstructing integer sets from their representation functions*, Electron. J. Combin. 11 (2004), R78.

[4] J. Cilleruelo and M. B. Nathanson, *Perfect difference sets constructed from Sidon sets*, Combinatorica 28 (2008), 401–414. `arXiv:math/0609244`.

[5] This note's accompanying code and data, `github.com/.../erdos_1194/`.